Java Programming Assignment

Data Types, Variables & Operators

Q1

Which of the following is NOT a primitive data type in Java?

A. int
B. boolean
C. String
D. char

Q2

What will be printed?

```
int x = 7;
int y = 2;
System.out.println(x / y);
```

A. 3
B. 3.5
C. 4
D. Compilation Error

Q3

Which declaration is valid?

A. float price = 19.99;
B. float price = 19.99f;
C. float price = "19.99";
D. float price = '19.99';

Q4

What is the value of x?

```
int x = 8;
x %= 3;
```

A. 0
B. 1
C. 2
D. 3

Q5

Which expression evaluates to false?

A. 5 == 5

B. 10 >= 5

C. 8 != 8

D. 7 < 9

Q6

Which operator has the highest precedence?

A. +

B. *

C. ==

D. &&

Q7

Which variable name follows Java naming conventions?

A. student-name

B. 2number

C. studentName

D. class

Q8

What is the output?

```
System.out.println(10 + 5 * 2);
```

Output=20

Q9

Predict the output.

```
int x = 5;
System.out.println(x++);
System.out.println(x);
```

Output 5,6

Q10

Predict the output.

```
int x = 5;
System.out.println(++x);
System.out.println(x);
```

Output 6,6

Q11

Determine the output.

```
int a = 10;
int b = 3;
System.out.println(a / b);
```

```
System.out.println(a % b);
```

Output 3,1

Q12

Predict the output.

```
boolean a = true;
boolean b = false;
System.out.println(a || b && false);
```

True

Q13

What will be printed?

```
int x = 2, y = 3, z = 4;
System.out.println(x + y * z - 1);
```

Outpu=13

Q14

```
int x = 10;
System.out.println(x++ + ++x);
```

Output 22

Q15

```
int x = 4;
System.out.println(x++ * 2 + --x);
```

Output 12

Q16

What is the problem?

```
int age = "20";
```

A. No error
B. Wrong data type assigned
C. Missing semicolon
D. Invalid variable name

Q17

Identify the error.

```
boolean passed = "true";
```

A. No error
B. "true" should be true
C. Use single quotes
D. boolean cannot store true/false

Q18

Will this code compile?

```
int x;
x = x + 5;
```

```
System.out.println(x);
```

A. Yes, it prints 5
B. Yes, it prints 0
C. Compilation Error
D. Runtime Error

Bonus

Without running the code, determine the output and explain your answer.

```
int a = 5;
int b = 10;
System.out.println(a++ + b-- + ++a + --b);
```